

SPATIAL MAPPING OF BOREHOLES & WATER SOURCES IN THE KGALAGADI DISTRICT , BOTSWANA

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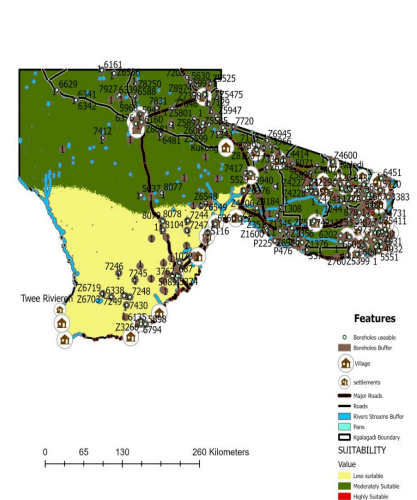
Module: Geographical Information Systems

Institution: BUAN — Botswana University of Agriculture

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STUDY AREA: KGALAGADI

SPATIAL ANALYSIS OF GROUNDWATER AND INFRASTRUCTURE IN KGALAGADI DISTRICT



PRESENTATION OUTLINE

01

Problem Statement

02

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PROBLEM STATEMENT

The Kgalagadi District covers a large stretch of the Kalahari Desert in southwestern Botswana. It is one of Botswana's most water-scarce regions with no permanent rivers or surface water bodies. Communities are almost entirely dependent on groundwater accessed through boreholes for their daily water needs. Botswana has 17,216 boreholes nationally, with approximately 2000 within Kgalagadi District.

Despite this critical dependence, there is limited spatial information on where these water sources are located which types exist (private or public) and which ones are functional,. This project seeks to address this gap by spatially mapping water sources across the Kgalagadi District.

17,216

Boreholes
total in Botswana

2,000

In Kgalagadi
after buffer analysis

**195**

Final Selected
Boreholes

**11.6K**

km² Area

OBJECTIVES

01

Map the spatial distribution of boreholes and water sources in Kgalagadi District

04

Map proximity of water sources to settlements and villages

02

Filter and identify boreholes with yield ≥ 10 and distance buffer ≥ 200

05

Perform weighted overlay to identify suitability zones (1=Less, 2=Moderate, 3=High)

03

Map the functionality and type of boreholes (RU, RC, LS, SU, DO, IR)

06

Visualize water coverage gaps and high density borehole areas

DATASETS

Dataset	Format	Purpose
Boreholes & Water Points	Shapefile / CSV	Primary water point locations
District Boundary	Shapefile	Study area boundary
Roads Network	Shapefile	Infrastructure layer
Rivers & Streams	Shapefile	Surface water network
Settlements	Shapefile	Settlement locations
Villages	Shapefile	Village locations
Pans	Shapefile	Natural water bodies
Digital Elevation Model	Raster (TIF)	Terrain & slope analysis

SELECTION CRITERIA

Dataset	Restriction / Criteria
Boreholes	Should be in use (public/community) with yield ≥ 10
Roads	Boreholes should be closer to existing roads
Villages & Settlements	Boreholes should be closer to populated areas
Digital Elevation (Slope)	Should be located on gentle slopes

BOREHOLE TYPES INCLUDED (via SQL Query)

RU

Rural

RC

Rural Community

LS

Livestock

SU

Sub Urban

DO

Domestic

IR

Irrigation

BOREHOLE TYPES EXCLUDED (via SQL Query)

UN

Unused

RE

Reserved

GN

Government

MO

Monitored

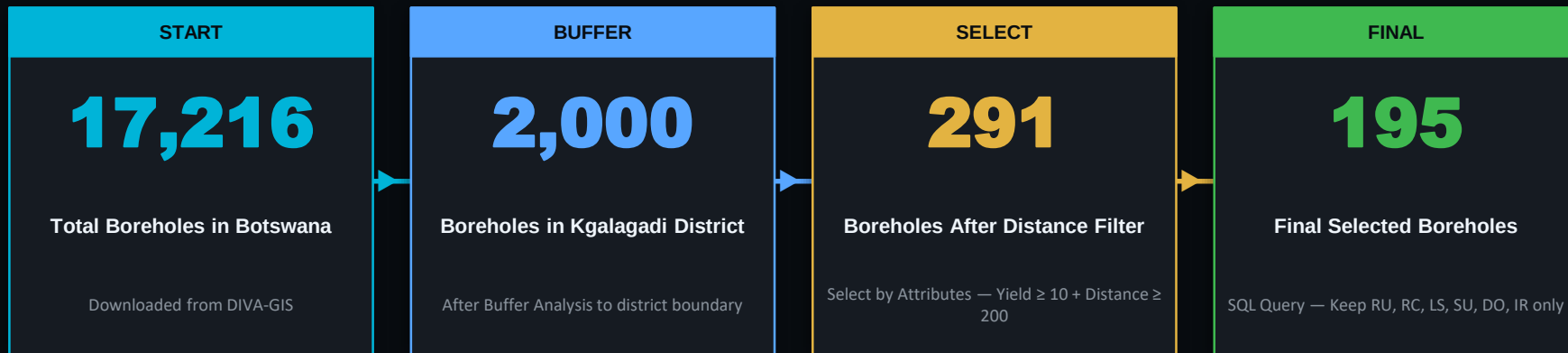
IN

Industrial

EX

Exploration

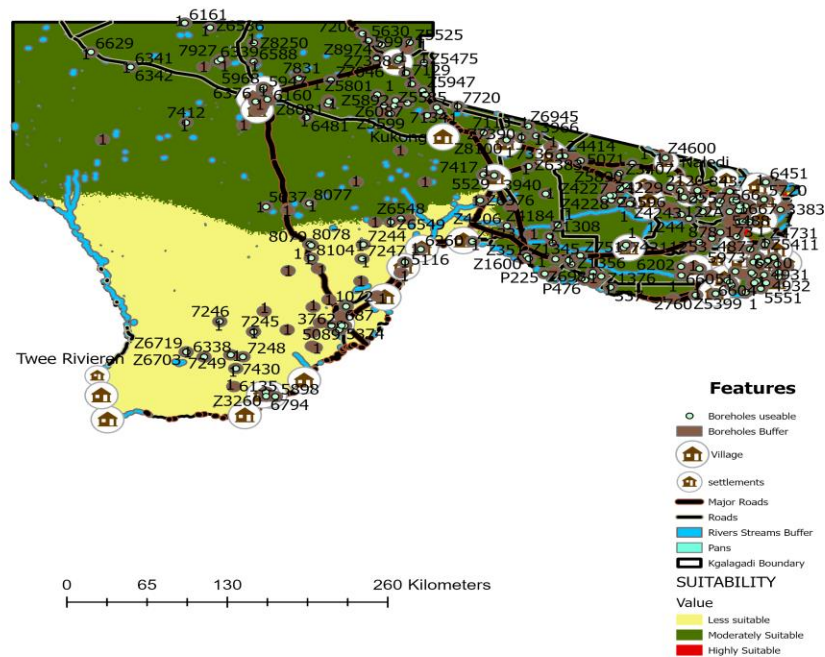
METHODOLOGY



TOOLS USED IN ARCGIS PRO

Step	Tool	Purpose	Result
1	Buffer (Analysis Tools)	Create buffer around Kgalagadi boundary	2,000 boreholes extracted
2	Select by Attributes	Filter Yield ≥ 10 + Distance ≥ 200	291 boreholes selected
3	SQL Query	Keep RU, RC, LS, SU, DO, IR types only	195 final boreholes
4	Clip	Clip all other layers to Kgalagadi boundary	All layers confined to ROI
5	Near (Analysis Tools)	Distance from settlements to boreholes	NEAR_DIST column added
6	Kernel Density	Borehole concentration hotspot map	Density raster output

N

A black arrow pointing upwards, indicating North.

WEIGHTED OVERLAY & SUITABILITY ANALYSIS

CLASS 1

1

Less Suitable

Low borehole density
Far from settlements & roads

CLASS 2

2

Moderately Suitable

Medium borehole density
Moderate proximity to services

CLASS 3

3

Highly Suitable

High density of boreholes
Close to settlements & roads
Gentle slopes

WEIGHTED OVERLAY INPUTS

Input Layer	Role in Analysis	Suitability Logic
195 Final Boreholes	Primary water point layer	Higher density = more suitable
Roads	Infrastructure access	Closer to roads = more suitable
Settlements & Villages	Population demand	Closer to settlements = more suitable
Slope (from DEM)	Terrain suitability	Gentler slopes = more suitable

BOREHOLE STATISTICS

2000

Total Boreholes

in Kgalagadi District

10+

Yield Threshold

mm discharge filter

200 km

Buffer Radius

access zone per borehole

GW

Water Type

All boreholes are groundwater

CONCLUSION & REFERENCES

CONCLUSIONS

- 17,216 boreholes exist in Botswana; 2,000 are in Kgalagadi District
- After filtering by yield ≥ 10 and distance ≥ 200 , only 291 boreholes remained
- SQL Query further refined to 195 usable public/community boreholes
- Weighted overlay revealed 3 suitability zones across the district
- Large parts of western Kgalagadi remain in the less suitable zone

RECOMMENDATIONS

- New boreholes should target less suitable western zones
- Priority drilling in areas more than 5km from existing boreholes
- Unused and reserved boreholes (UN, RE) should be rehabilitated
- Regular spatial monitoring of borehole yield and functionality

REFERENCES

DIVA-GIS (2024). Free Spatial Data for Botswana. diva-gis.org

THANK YOU
